This equation represents the basic relationship for the production of radionuclides; in a reactor k is calculated according to (17.12) but for accelerator irradiation, where the current and not the flux is known, (17.13) is normally used.

Let us again take the production of radioactive sodium by reactor irradiation and cyclotron bombardment as examples of the application of these equations. Na_2CO_3 (M = 106) is irradiated in a reactor to produce ^{24}Na, which has a half-life of 14.66 h and emits β^- and γ-rays. The reaction cross-section of ^{23}Na (100% in natural sodium) is 0.53 b for thermal neutrons. For a 5 g sample, 60 h irradiation time, and a thermal flux of 10^{12} n cm^{-2} s^{-1}, the activity at the end of the irradiation time will be:

$$A = 10^{12} \times 0.53 \times 10^{-24} \times 2 \times 5 \times 6.02 \times 10^{23}\left(1 - e^{-0.693 \times 60/15}\right)/106$$
$$= 2.8 \times 10^{10}\,\text{Bq} \tag{17.15}$$

^{24}Na may be produced through accelerator irradiation of magnesium by the reaction ^{26}Mg (d,α) ^{24}Na. With 22 MeV D^+ ions the reaction cross-section is about 25 mb. For a 0.1 mm thick magnesium foil (M = 24.3, $\rho = 1.74$ g cm^{-3}) with an area larger than the projectile beam, and a 2 h irradiation at 100 μA, the activity of ^{24}Na produced (natural magnesium contains 11.0% ^{26}Mg) at end of irradiation is:

$$A = 6.24 \times 10^{18} \times 100 \times 10^{-6} \times 25 \times 10^{-27}\left(0.11 \times (1.74/24.3)\right.$$
$$\left. \times\, 6.02 \times 10^{23} \times 0.01\right) \times \left(1 - e^{-0.693 \times 2/15}\right) = 6.53 \times 10^{7}\text{Bq} \tag{17.16}$$

17.3. SECOND-ORDER REACTIONS

For large cross-sections, short half-lives, and long bombardments, second-order capture products may be formed. If the first-order product is radioactive, then its concentration at any time is dependent on (i) the decay constant, (ii) the cross-section for production of the second-order product, as well as (iii) the cross-section for its own production. These possibilities lead to a scheme such as Figure 17.2. For simplicity, we may assume that the induced nuclear transformations only involve single-neutron capture, and that only β^- decay occurs. However, it should be obvious that the scheme in Figure 17.2, with suitable additions, is applicable to all kinds of nuclear formation and decay reactions.

For the (n,γ) β^- case the upper horizontal row of Figure 17.2 represents the successive formation of higher isotopes of the target element (the constant Z-chain) and the vertical rows the isobaric decay chains of each of these isotopes (the constant A-chains). The first of these two rows is indicated by heavy arrows. Chains which involve both induced transformations and radioactive decay play a central role in theories about the

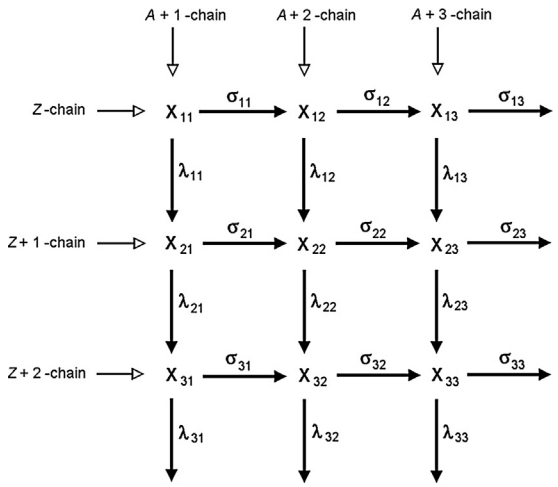


Figure 17.2 Formation of higher nuclides through multiple neutron capture and their associated decay chains.

formation of the elements in the universe, in the thermonuclear reactions in the stars (Ch. 12), and in the synthesis of transuranium elements (Ch. 14).

For each nuclide in Figure 17.2 we can write a differential expression for the change in concentration with time as a function of the irradiation and the radioactive decay:

$$\text{For } X_{11} : dN_{11}/dt = -(\varphi\sigma_{11} + \lambda_{11})N_{11} = \Lambda_{11}N_{11} \tag{17.17}$$

$$X_{21} : dN_{21}/dt = \lambda_{11}N_{11} - (\varphi\sigma_{21} + \lambda_{21})N_{21} = \Lambda^*_{11}N_{11} - \Lambda_{21}N_{21} \tag{17.18}$$

$$X_{31} : dN_{31}/dt = \lambda_{21}N_{21} - (\varphi\sigma_{31} + \lambda_{31})N_{31} = \Lambda^*_{21}N_{21} - \Lambda_{31}N_{31} \tag{17.19}$$

$$\ldots \qquad \ldots$$

$$X_{12} : dN_{12}/dt = \varphi\sigma_{11}N_{11} - (\varphi\sigma_{12} + \lambda_{12})N_{12} = \Lambda^*_{11}N_{11} - \Lambda_{12}N_{12} \tag{17.20}$$

$$X_{13} : dN_{13}/dt = \varphi\sigma_{12}N_{12} - (\varphi\sigma_{13} + \lambda_{13})N_{13} = \Lambda^*_{12}N_{12} - \Lambda_{13}N_{13} \tag{17.21}$$

$$\ldots \qquad \ldots$$

$$X_{22} : dN_{22}/dt = \lambda_{12}N_{12} + \varphi\sigma_{21}N_{21} - (\varphi\sigma_{22} + \lambda_{22})N_{22} \tag{17.22}$$

Let us now consider three different cases:

Case (i): *Successive radioactive decay.* This occurs after formation of the first radioactive (parent) member X_{11}. Thus we only consider the left vertical chain, and assume σ_{11}, σ_{21}, etc. $= 0$. In this $A + 1$ chain the second index is constant, and we omit it for case (i). This

case is valid for all natural decay chains and for fission product decay chains. Each member of the decay chain can be described by the differential equation

$$dN_i/dt = \lambda_{i-1}N_{i-1} - \lambda_i N_i \ (i > 1) \tag{17.23}$$

We have already solved this relation in §17.2 for the formation of X_1 (the radioactive mother), and in §5.15 we showed how the amount of X_2 (the radioactive daughter) varies with time.

The general solution to the case with many successive decays is usually referred to as the *Bateman equations* (H. Bateman 1910):

$$N_n(t) = \lambda_1\lambda_2 \ldots \lambda_{n-1}\ N_1^0 \sum_1^n C_i e^{-\lambda_i t} \tag{17.24}$$

$N_n\ (t)$ is the number of atoms for the nth species after time t, N_1^0 is the number of atoms of X_1 at $t = 0$, t is the *decay time*, and

$$C_i = \prod_{j=1}^{j=n} \left(\lambda_j - \lambda_i\right)^{-1} \ \text{for}\ j \neq i \tag{17.25}$$

For $i = 1$, eqn. (5.58) for simple radioactive decay is obtained. It is assumed that $N_i^0 = 0$ for $(i > 1)$. If that is not the case, additional chains, starting with each $N_i > 0$, require a new series of Bateman equations which are added together. For $i = 3$ and $N_2^0 = N_3^0 = 0$ at $t = 0$ one obtains

$$N_3(t) = \lambda_1\lambda_2\, N_1^0 \left(C_1\, e^{-\lambda_1 t} + C_2\, e^{-\lambda_2 t} + C_3\, e^{-\lambda_3 t}\right) \tag{17.26}$$

where

$$C_1 = (\lambda_2 - \lambda_1)^{-1}\ (\lambda_3 - \lambda_1)^{-1} \tag{17.27}$$

$$C_2 = (\lambda_1 - \lambda_2)^{-1}\ (\lambda_3 - \lambda_2)^{-1} \tag{17.28}$$

$$C_3 = (\lambda_1 - \lambda_3)^{-1}\ (\lambda_2 - \lambda_3)^{-1} \tag{17.29}$$

Case (ii): *Successive neutron capture*. This is the upper horizontal row in Figure 17.2 (the constant Z-chain). Because the first index is constant in this chain, we omit it for case (ii). We assume λ_{11}, λ_{12}, etc. $= 0$. This is a valid approximation often used for the formation of transuranium elements in a nuclear reactor, which are long-lived in relation to the time of irradiation and observation. For each member of the formation chain we may write

$$dN_i/dt = \varphi\sigma_{i-1}N_{i-1} - \varphi\sigma_i N_i \tag{17.30}$$

By replacing λ_i with $\varphi\sigma_i$ this equation becomes identical with (17.23). Thus the solution of case (ii) is obtained by using the Bateman eqn. (17.24) and (17.25). It should be noted that in case (ii) t is the *irradiation time* (t_{irr}).

Case (iii): *Combined induced transformation and radioactive decay.* These combined effects can be taken into account by using the Bateman equations with some modifications, as developed by Rubinson.

We introduce the abbreviations

$$\Lambda = \varphi\sigma + \lambda \tag{17.31}$$

and

$$\Lambda^* = \varphi\sigma^* + \lambda^* \tag{17.32}$$

Λ_i is referred to as the disappearance constant of nuclide i, while Λ_i^* is the partial formation constant, and σ^* and λ^* the partial reaction cross-sections and decay constants for nuclide $i + 1$; σ, σ^*, λ and λ^* may be equal to zero in some cases. The exact meaning of Λ and Λ^* is explained below with examples. Introducing these abbreviations for the formation of the $X_{i\ 1}$ species of the constant $(A + 1)$-chain, and dropping the second index, which is =1 in the left vertical row in Figure 17.2, we obtain the equation:

$$dN_i/dt = \lambda_{i-1}N_{i-1} - \Lambda_i N_i = \Lambda_{i-1}^* N_{i-1} - \Lambda_i N_i \tag{17.33}$$

Similarly, we find for the formation of the $X_{1\ i}$ species (the constant Z-chain)

$$dN_i/dt = \varphi\sigma_{i-1}N_{i-1} - \Lambda_i N_i = \Lambda_{i-1}^* N_{i-1} - \Lambda_i N_i \tag{17.34}$$

where the first index (=1) has been dropped for simplicity. These two equations turn out to be identical, provided the partial formation constant Λ^* (17.32) for each nuclide (i.e. each index set) is used with the correct σ^* and λ^*; in (17.33) $\sigma^* = 0$, and in (17.34) $\lambda^* = 0$. With this reservation in mind the Bateman equations become

$$N_n(t) = \Lambda_1\Lambda\ldots\Lambda_{n-1}^* N_1^0 \sum_1^n C_i e^{-\Lambda_i t} \tag{17.35}$$

where

$$C_i = \prod_{j=1}^{j=n} \left(\Lambda_j - \Lambda_i\right)^{-1} (j \neq i) \tag{17.36}$$

The time t is the irradiation time t_{irr}.

The application of this expression and the use of Λ^* are best illustrated with a practical example.

If 1.0 mg of Tb_2O_3 (M = 366) is bombarded for 30 days at a flux of thermal neutrons of 10^{14} cm^{-2} s^{-1}, the following processes occur:

$$\begin{array}{ccccc} {}^{159}\mathrm{Tb} & \xrightarrow[\sigma_{159}\ 25.5\ \mathrm{b}]{(n,\gamma)} & {}^{160}\mathrm{Tb} & \xrightarrow[\sigma_{160}\ 525\ \mathrm{b}]{(n,\gamma)} & {}^{161}\mathrm{Tb} \\ & & \lambda_{160}\ \Big|\ t_{1/2}\,72.1\ \mathrm{d} & & \lambda_{161}\ \Big|\ t_{1/2}\ 6.90\ \mathrm{d} \\ & & \downarrow & & \downarrow \\ & & {}^{160}\mathrm{Dy\ (stable)} & & {}^{161}\mathrm{Dy\ (stable)} \end{array} \tag{17.37}$$

The appropriate equation for the formation of ^{161}Tb (cf. also (17.26)) is:

$$\begin{aligned} N_{161} = \Lambda_{159}\Lambda^*_{160}N^0_{159}\big[&\mathrm{e}^{-\Lambda_{159}\,t}/\{(\Lambda_{160}-\Lambda_{159})(\Lambda_{161}-\Lambda_{159})\} \\ &+\mathrm{e}^{-\Lambda_{160}t}/\{(\Lambda_{159}-\Lambda_{160})(\Lambda_{161}-\Lambda_{160})\} \\ &+\mathrm{e}^{-\Lambda_{161}t}/\{(\Lambda_{159}-\Lambda_{161})(\Lambda_{160}-\Lambda_{161})\}\big] \end{aligned} \tag{17.38}$$

By considering the different reaction paths, we obtain

$$(\Lambda_1 =) : \Lambda_{159} = \varphi\sigma_{159} = 10^{14}\times 25.5\times 10^{-24} = 2.55\times 10^{-9}\,\mathrm{s}^{-1} \tag{17.39}$$

$$\begin{aligned} (\Lambda_2 =) : \Lambda_{160} &= \varphi\sigma_{160} + \lambda_{160} = 10^{14}\times 525\times 10^{-24} \\ &\quad + 0.693/(72.1\times 24\times 3600) = 1.64\times 10^{-7}\,\mathrm{s}^{-1} \end{aligned} \tag{17.40}$$

$$(\Lambda_3 =) : \Lambda_{161} = \lambda_{161} = 0.693/(6.9\times 24\times 3600) = 1.162\times 10^{-6}\,\mathrm{s}^{-1} \tag{17.41}$$

$$(\Lambda^*_3 =) : \Lambda^*_{160} = \varphi\sigma_{160} = 10^{14}\times 525\times 10^{-24} = 5.25\times 10^{-8}\,\mathrm{s}^{-1} \tag{17.42}$$

$$\left(N^0_1 =\right) : N^*_{159} = (2/366)\times 6.02\times 10^{23}\times 10^{-3} = 3.29\times 10^{18}\ \text{atoms} \tag{17.43}$$

$$(t =) : t = t_{\mathrm{irr}} = 30\times 24\times 3600 = 2.59\times 10^{6}\,\mathrm{s} \tag{17.44}$$

Note that Λ^*_{159} is the rate of formation of ^{160}Tb, and Λ^*_{160} the rate of formation of ^{161}Tb, while Λ_{159}, Λ_{160} and Λ_{161} are the rates of disappearance of ^{159}Tb, ^{160}Tb and ^{161}Tb. Introducing the experimental values yields $N_{161} = 5.68\times 10^{14}$ atoms. The radioactive decay is $A_{161} = \lambda_{161}\,N_{161} = 1.162\times 10^{-6}\times 5.68\times 10^{14} = 6.60\times 10^{8}$ Bq or 4×10^{10} dpm.

Alternatively to this approach Laplace transforms can be efficiently used to directly solve the system of differential equations shown in equation 17.17 to 17.22. By using this technique a full flexibility is achieved when it comes to the problem formulation and which events to take into account. Each equation (17.17 to 17.22) is then transformed

into Laplace space where a differential equation becomes a linear equation. Thus in Laplace space equation 17.18 becomes:

$$sN_{21} - \widehat{L}N_{21}^{0} = \lambda_{11}\widehat{L}N_{11} - (\varphi\sigma_{21} + \lambda_{21})\,\widehat{L}N_{21} \tag{17.45}$$

The system of now linear equations is solved with respect to the desired nuclide and the solution is transferred back into normal space if the inverse transform is known.

17.4. TARGET CONSIDERATIONS

The success of an irradiation experiment depends to a large extent on the target considerations. In many nuclear experiments as much time has to be devoted to the target preparation as to the rest of the experiment. If the purpose is to produce a radionuclide for a simple tracer experiment, the consideration in this section may be sufficient. If, however, the requirements are a product of extreme purity, very high specific activity, or very short half-life, special techniques must be used, which are discussed in §§17.5–17.7. The production of atoms of the heavy elements also require fast and efficient production and separation methods.

17.4.1. Physical properties

In thin targets the flux and energy of the projectile do not change during passage through the target and calculation of e.g. production rates are straight forward by applying e.g. equation 17.14 using k from 17.12. If, however the projectile energy is considerably changed (in a thick target) it is more complicated to calculate the yield as the variation of energy with distance in the target and the variation of cross section with energy must be taken into account.

The exact specification of when a target is thin depends on the experiment; in most cases thin means a target with a surface weight of not more than a few mg/cm^2 or a thickness of a few μm or less. The *surface weight* (or *linear density*) is $x\rho$, where x is the target thickness and ρ its density. Thin targets are usually used in accelerator irradiations and are made of a solid material, either metal or compound. In order to have sufficient mechanical strength the target is usually supported on a *backing material* such as aluminum. The backing material may be part of a solid target holder which has provisions for cooling to dissipate the heat developed in the bombardment. Thin targets are fixed on the backing material through electrolytic precipitation, vacuum deposition, mass spectrometric collection (for isotopically pure targets), or a number of other techniques.

In reactor irradiations it is more common to use thicker targets in order to obtain higher yields. In traversing the target the effective flux (i.e. $\varphi\sigma$) does not decrease as rapidly for neutrons as for charged particles and, therefore, thicker targets can still be considered "thin" in neutron irradiations. However, if the product σN is large, the neutron flux may decrease appreciably at the center of the target, especially at the neutron energies where cross sections are large.